Next you'll define the subnet for the proxy.

5.5.2 Defining the public proxy subnet

The EC2 instance running the proxy needs to be reachable via the internet. To achieve that, you'll need to complete the next four steps:

1. Create a subnet spanning a subsection of the IP address range assigned to the VPC.
2. Create a route table, and attach it to the subnet.
3. Add the `0.0.0.0/0` route pointing to the internet gateway to the route table.
4. Create an NACL, and attach it to the subnet.

To allow traffic from the internet to the proxy machine and from the proxy to the backend servers, you'll need the following NACL rules, shown in the next code snippet:

- Internet to proxy: HTTP from 0.0.0.0/0 to 10.0.1.0/24 is allowed.
- Proxy to backend: HTTP from 10.0.1.0/24 to 10.0.2.0/24 is allowed.

```
SubnetPublicProxy:
  Type: 'AWS::EC2::Subnet'
  Properties:
    AvailabilityZone: !Select [0, !GetAZs '']     ①
    CidrBlock: '10.0.1.0/24'                      ②
    MapPublicIpOnLaunch: true
    VpcId: !Ref VPC
    Tags:
    - Key: Name
      Value: 'Public Proxy'
RouteTablePublicProxy:                            ③
  Type: 'AWS::EC2::RouteTable'
  Properties:
    VpcId: !Ref VPC
```